

# Ann Mathew

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## OVERVIEW

Software engineer with interdisciplinary full-stack mobile and AI-integrated development experience. Mid-level professional who contributed to building a 4M+ Amazon seller database and analyzed 37,000+ patient medication records to improve EHR accuracy. Proven ability to deliver data-driven solutions and scalable applications. Seeking to apply technical and analytical skills to drive product innovation and improve user outcomes.

## EDUCATION

### Vanderbilt University | School of Engineering

Nashville, TN

*Bachelor of Science; Majors in Computer Science, Climate & Environmental Studies*

*August 2022 – June 2026*

- [Portfolio Site](#): [built-by-ann.github.io](https://built-by-ann.github.io) | [GitHub](https://github.com/built-by-ann): [github.com/built-by-ann](https://github.com/built-by-ann)
- **Relevant Coursework**: Algorithms, Data Structures, Operating Systems, Database Management Systems, Programming Languages, Intermediate Software Design, Senior Projects in Web Systems, Projects in UX/UI, Artificial Intelligence
- **Study Abroad**: Siena, Italy (Spring 2025)

## SKILLS

- **Languages**: Python, JavaScript/TypeScript, SQL, Java, C/C++, HTML/CSS
- **Frameworks & Technologies**: React Native, Node.js, Fastify, MongoDB, REST APIs, LLM Integration, Flask, Firebase
- **Developer Tools**: Git/GitHub, Expo, Figma, Agile Development, Claude Code, Tableau, PowerBI, QGIS
- **Data & ML Tools**: Pandas, GeoPandas, scikit-learn, PyTorch, TensorFlow, NumPy, Jupyter

## PROJECTS

### MorningCall | AI-Powered Full-Stack Mobile Application

Nashville, TN

*Full-Stack Developer for Projects in Web Systems @ VU*

*January 2026 – May 2026*

- Developed an AI-powered mobile productivity assistant using React Native, Fastify, MongoDB, and Firebase within a collaborative monorepo architecture
- Built full-stack features including habit tracking, task management, conversational AI, voice integration, and responsive mobile interface

### Herb–Disease Association Prediction | Environmental Pharmacology Network Analysis

Nashville, TN

*Researcher for Network Analysis in Healthcare @ VU*

*August 2025 – December 2025*

- Developed an augmented herb-disease association prediction framework by integrating environmental similarity data into the HDAPM-NCP network analysis model using Python scientific computing workflow
- Engineered climate similarity kernels using GBIF occurrence records and Köppen-Geiger climate classifications, then evaluated fusion strategies using AUROC and AUPR performance metrics
- Built a reproducible environmental data integration pipeline and analyzed climate-driven restructuring of herb similarity network topology

### Land Conservation Risk Modeling | Geospatial Environmental Machine Learning

Nashville, TN

*Machine Learning Project Developer for Vanderbilt Data Science @ VU*

*November 2023 – May 2025*

- Developed machine learning models using Landsat satellite imagery and geospatial environmental data to identify high-risk land areas for conservation in collaboration with the Land Trust for Tennessee
- Produced geospatial raster datasets, methodology documentation, and an interactive web application for environmental data visualization and conservation-focused decision support

## PROFESSIONAL EXPERIENCE

### Vanderbilt University Medical Center | Department of Biomedical Informatics

Nashville, TN

*Research Intern*

*May 2024 – August 2024*

- Collaborated with Dr. Allison McCoy (PhD) to assess gaps in EHR medication records, contributing to recommendations for improving the accuracy of medication documentation in electronic health records, specifically when medications are no longer being taken.
- Conducted a comprehensive analysis of medication discontinuation practices through evaluation of 37,000+ patient encounters and interviewing physicians across various specialties, uncovering key inconsistencies and inaccuracies in medication documentation in the EHR.

### TrackSellers | Amazon Marketplace Analytics Startup

Remote

*Data Science Intern*

*May 2022 – August 2022*

- Built a database of 4M+ Amazon sellers by web-scraping with Python (Scrapy, XPath) and storing the data in MySQL, producing detailed seller profiles and rankings that improved partnership targeting within the Amazon ecosystem
- Facilitated connections between suppliers and sellers through tailored ranking reports and insights on competition and market niches.